



ACADEMIC REGULATIONS & SYLLABUS

(Choice Based Credit System)

Faculty of Technology & Engineering
Chandubhai S. Patel Institute of Technology
Department of Mechanical Engineering
Bachelor of Technology Programme
(Fourth Year Mechanical Engineering)





CHAROTAR UNIVERSITY OF SCIENCE AND
TECHNOLOGY

Faculty of Technology and Engineering



ACADEMIC REGULATIONS

Bachelor of Technology (Mechanical Engineering) Programme
(Choice Based Credit System)

Charotar University of Science and Technology (CHARUSAT)

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Academic Year 2024-25

FACULTY OF TECHNOLOGY AND ENGINEERING

ACADEMIC REGULATIONS

Bachelor of Technology Programmes

Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1) System of Education

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Postgraduate levels. Each semester will have at least 90 working days duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2) Duration of Programme

i)	Undergraduate programme	(B.Tech)
	Minimum	8 semesters (4 academic years)
	Maximum	16 semesters (8 academic years)

3) Eligibility for admissions

As enacted by Govt. of Gujarat from time to time.

4) Mode of admissions

As enacted by Govt. of Gujarat from time to time.

5) Programme structure and Credits

As per annexure – 1 attached

6) Attendance

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be 80%.

7) Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment. The continuous assessment will be conducted by the respective department/ institute.

7.1.2 Final end-semester examination by the University through written paper or practical test or oral test or presentation by the student or combination of these.

7.1.3 The weightages of continuous assessment and end-semester University examination in overall assessment shall depend on individual course as approved by Academic Council through Board of Studies.

7.1.4 The performance of candidate in continuous assessment and end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.

7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2 Performance in continuous assessment and end-semester University Examinations

7.2.1 Minimum performance with respect to continuous assessment as well as end-semester university examination will be an important consideration for passing a

course. Details of minimum percentage of marks to be obtained in the examinations are as follows:

Minimum percentage marks to be obtained in end-semester University examination (for applicable courses)	Minimum overall percentage marks to be obtained in each course
40%	45%

7.2.2 If a candidate obtains minimum required percentage of marks in end-semester University examination in an applicable course but fails to obtain minimum required overall percentage of marks, he/she has to repeat the examination till the minimum required overall percentage of marks are obtained.

8) Grade Point System

8.1. The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table 1: Grading Point System (UG)

Range of Marks (%)	≥80	≥73 <80	≥66 <73	≥60 <66	≥55 <60	≥50 <55	≥45 <50	<45
Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	09	08	07	06	05	04	00

8.2. The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

$$(i) \quad SGPA = \frac{\sum C_i G_i}{\sum C_i} \quad \text{where } C_i \text{ is the number of credits of course } i$$

G_i is the Grade Point for the course i

and $i = 1$ to n , n = number of courses in the

semester

- (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters
up to which CGPA is computed.

9) Award of Class

- ☞ The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
First Class with Distinction	$CGPA \geq 7.5 \text{ \& } \leq 10.0$
First class	$CGPA \geq 6.0 \text{ \& } < 7.5$
Second Class	$CGPA \geq 5.0 \text{ \& } < 6.0$
Pass Class	$CGPA < 5.0$

Grade sheets of only the final semester shall indicate the class. In case of all the other semesters, it will simply indicate as Pass / Fail.

9.1. Maximum duration allowed for Completion of a programme

- ☞ Maximum duration to allow for completion of a particular programme shall not be more than twice the normal duration of the respective programme. For example, a 6-Semester programme should be completed within not more than 12 semesters.

10) Detention Criteria

- ☞ No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.
- ☞ A Student will not be allowed to move to third year if he/she has not cleared all the courses of first year.
- ☞ A student will not be allowed to move to fourth year if he/she has not cleared all the courses of first and second year.

11) Transcript

- ☞ A transcript issued to the student at the time of leaving the university will contain a consolidated record of all the courses taken by him / her, grades obtained and the final CGPA.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
(CHARUSAT)

FACULTY OF TECHNOLOGY & ENGINEERING
(FTE)

CHOICE BASED CREDIT SYSTEM

FOR

BACHELOR OF TECHNOLOGY & ENGINEERING

CHOICE BASED CREDIT SYSTEM

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 4 types of courses: Foundation courses, Core courses, Elective courses and Non-credit (audit) courses.

1.1. Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

1.2. Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme. Environmental science will be a compulsory University core for all Undergraduate Programmes.

B. Programme Core courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

1.3. Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialised or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes. As a general guideline, Programme should incorporate 2 University Electives of 2 credits each (total 4 credits).

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

D. Cluster Elective Course (CE):

An 'Elective Course' is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg. Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

1.4. Non Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will be reflected in Student's Grade Sheet but the grade of the course will not be consider to calculate SGPA and CGPA. Attendance and Course Assessment is compulsory for Non Credit Courses.

1.5. Medium of Instruction

The Medium of Instruction will be English.

Department of Mechanical Engineering – Chandubhai S Patel Institute of Technology

Vision

To be recognized as a center for outstanding education and research in field of mechanical engineering.

Mission

To produce well qualified engineers, who are innovative, contributors to their profession by catering to diverse societal needs.

Program Educational Objectives:

The objectives of the Mechanical Engineering Undergraduate Program are to produce graduate who can:

1. Apply their knowledge and expertise to find solutions of the problems in core and allied areas of engineering, exhibiting creativity and entrepreneurial skills.
2. Conduct themselves in an ethical and professional manner, exhibiting effective skills in communication, management, teamwork and leadership.
3. Continue their intellectual development through self-study and continuing education, practicing life-long learning.

Program Outcomes: Mechanical Engineering

- PO1 **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2 **Problem analysis:** Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3 **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO4 **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5 **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and

modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

- PO6 **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO7 **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- PO8 **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9 **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10 **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11 **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- PO12 **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes: Mechanical Engineering

- PSO1 The Mechanical Engineering graduates will be able to analyse, design, and evaluate the performance of mechanical components and systems by using various tools.
- PSO2 The Mechanical Engineering graduates will be able to plan, automate and evaluate manufacturing and system processes.

TEACHING & EXAMINATION SCHEME

B. TECH. PROGRAMME IN MECHANICAL ENGINEERING

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)							(w. e. f. June 2023)				
TEACHING & EXAMINATION SCHEME FOR FOURTH YEAR B. TECH. PROGRAMME IN MECHANICAL ENGINEERING											
Level	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
Sem-7	ME443	Operation Research	3	0	3	3	30	70	0	0	100
	ME453	Design of Transmission Systems	3	0	3	3	30	70	0	0	100
	ME447	Power Plant Engineering	3	0	3	3	30	70	0	0	100
	ME448	Computer Aided Manufacturing	3	2	5	4	30	70	25	25	150
	ME4XX	Programme Elective - III	3	0	3	3	30	70	0	0	100
	ME4XX	Programme Elective - IV	3	2	5	4	30	70	25	25	150
	ME452	Summer Internship - II *	-	-	-	3	0	0	75	75	150
Sem-8		*(Internship after the 6th sem.)			22	23					850
	ME456	Major Project	0	28	28	14	0	0	200	200	400

Programme Electives:

Sr. No.	Course Nature	Course Code	Course Title
1	Programme Elective- III	ME492	Heating, Ventilation & Air Conditioning
2		ME485	Advanced Materials
3		ME484	Design of Heat Exchanger
4		ME483	Industrial Automation & Robotics
5		ME475	Product Design & Value Engineering
6		ME487	Project Management
7		ME488	Mechanical Estimation & Costing
8		ME 493	Finite Element Methods
9		ME490	Advanced Machine Design
10		ME491	Artificial Intelligence
11	Programme Elective- IV	ME495	Thermal System Design
12		ME477	Optimization Techniques
13		ME478	Industrial Tribology
14		ME479	Surface Engineering
15		ME480	Advanced Manufacturing Technology
16		ME481	Computational Fluid Dynamics & Heat Transfer
17		ME494	Industry 4.0 & Industrial IoT

B. Tech. (Mechanical Engineering) Programme

SYLLABI

(Semester – 7)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ME – 443 OPERATION RESEARCH

7th Semester and 4th Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Basics of operation research	02
2.	Linear programming	16
3.	Transportation models	04
4.	Assignment models	03
5.	Games theory	03
6.	Replacement models	04
7.	Queuing theory	04
8.	Network analysis	04
9.	Dynamic programming	05

Total hours (Theory): 45

Total: 45

B. Detailed Syllabus:

1	Basics of Operations Research	02 Hours	04%
1.1	Definition, characteristics, phases, scope, and limitations of OR		
2	Linear Programming Methods	16 Hours	36%
2.1	Formulation of problem, Graphical method, Simplex method		
2.2	Big M method, Two phase method, Degeneracy		
2.3	Unboundedness, Infeasibility, Cycling, Duality		
2.4	Solution of Linear Equations by Simplex method		

- | | | | |
|----------|---|-----------------|------------|
| 3 | Transportation Models | 04 Hours | 9% |
| 3.1 | North West Corner rule, Least cost method, Vogel's approximation method | | |
| 3.2 | Degeneracy in transportation problem, steppingstone method, Modified Distribution Method (MODI's method) | | |
| 3.3 | Unbalanced supply and demand, profit maximization problem, prohibited transportation routes, transshipment problems | | |
| 3.4 | Shortest route problem, minimum spanning tree problem, and maximum flow problem | | |
| 4 | Assignment Models | 03 Hours | 6% |
| 4.1 | Hungarian method for solution, non-square matrix, restriction on assignments, Maximization problem, Traveling Salesman Problem (TSP) | | |
| 5 | Game Theory | 03 Hours | 6% |
| 5.1 | Terms used in game theory, Two-person zero sum games, pure strategy, matrix reduction by dominance, mixed strategies (2x2, 2xn, mx2, 3x3 games) | | |
| 5.2 | Algebraic, arithmetic, and graphical method. | | |
| 6 | Replacement Models | 04 Hours | 09% |
| 6.1 | Introduction, Reasons for Replacement Equipment, Emergence of Equipment Replacement | | |
| 6.2 | Replacement Models, MAPI Method, Sudden Failure Items, Failure Trees | | |
| 7 | Queuing Theory | 04 Hours | 09% |
| 7.1 | Elements of Queuing System, Characteristics of Waiting Lines, Service Discipline | | |
| 7.2 | Service Mechanism | | |
| 8 | Network Analysis | 04 Hours | 10% |
| 8.1 | Terms used in network analysis, Network or arrow diagram, Fulkerson's rule | | |
| 8.2 | Programme evaluation and review technique (PERT), Critical path method (CPM), Crashing of network | | |
| 9 | Dynamic Programming | 05 Hours | 11% |
| 9.1 | Introduction, Bellman's principle of Optimality, solution of problems with finite | | |

number of stages

9.2 Solution of LPP by dynamic programming

C. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures behaviour will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Explain various Operations Research techniques and their applications in real-life problems.

CO2: Select and apply tools and techniques of operation research to find appropriate solutions to decision making problems.

CO3: Solve linear and dynamic programming problems.

CO4: Built and solve the transportation and assignment models.

CO5: Solve the replacement and queuing problems.

CO6: Analyze and solve the network and game theory problems.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	-	-	-	-	-	-	-	1	1	-	-	-	-
CO2	1	2	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	3	1	-	-	-	-	-	-	-	-	-	-	-
CO4	-	3	1	-	-	-	-	-	-	-	-	-	-	-
CO5	-	3	1	-	-	-	-	-	-	-	-	-	-	-
CO6	-	3	1	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Sharma J.K., “Operations Research: Theory and Applications”, Trinity Laxmi Publications.
2. Hira D. S. and Gupta P. K., “Operations Research”, S. Chand Publications.
3. Verma A. P., “Operation Research”, S K Kataria and Sons, New Delhi.
4. Sen Rathindra P., “Operation Research: Algorithms and applications”, Prentice Hall of India.
5. Vohra N. D., “Quantitative Techniques in Management”, Tata McGraw Hill, New Delhi.

Reference books:

1. Pannerselvam R., “Operations Research”, Prentice Hall of India, New Delhi.
2. Sharma J. K., “Operations Research: Problems and Solutions”, Macmillan India Ltd.
3. Wagner, “Principles of Operations Research: With Applications to Managerial Decisions”, Prentice Hall of India, New Delhi.
4. Taha Hamdy, “Operations Research: An Introduction”, Prentice Hall India.
5. Hillier and Lieberman, “Introduction to Operations Research”, Tata McGraw Hill.
6. Natarajan A. M., Balasubramani P. and Tamilarasi A., “Operations Research”, Pearson Education.
7. Winston and Goldberg, “Operations Research: Applications and Algorithms”, Thomson Brooks.
8. Maurice Saseini, Arhur Yaspan and Lawrence Friedman, “Operations Research: Methods and Problems”, Wiley. Hans-Joachim. Mittag, Horst Rinne Statistical Methods of Quality assurance CRC Press, 1993

Other Material:

1. Programming Languages: MATLAB
2. Science Direct Journal (<http://www.sciencedirect.com>)
3. IEEE transactions (<http://ieeexplore.ieee.org>)
4. Mechanical Engg. (Inst. of Engineers) <http://www.ieindia.org>

ME453 DESIGN OF TRANSMISSION SYSTEM

7th Semester and 4th Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Min. no of Hrs.
1.	Gear Design	09
2.	Design of Flexible Elements	06
3.	Design of Gear Box	08
4.	Design of Sliding Contact Bearings	07
5.	Design of Rolling Contact Bearings	07
6.	Design of I.C. Engine Components	08

Total Hours (Theory): 45

Total Hours: 45

B. Detailed Syllabus:

- | | | | |
|-----|---|-----------------|------------|
| 1 | Gear Design | 09 Hours | 20% |
| 1.1 | Introduction to gear terminology, law of gearing, interference and undercutting materials | | |
| 1.2 | Gear tooth failures, basic theory of gear design- Lewis equation, beam strength, strength, etc. | | |
| 1.3 | Design of Spur Gear, Helical Gear, Bevel Gear, Worm Gear | | |
| 2 | Design of Flexible Elements | 06 Hours | 12% |
| | Belt Drives: Classification of belt drives, Mathematical Relationship of belt tension, | | |
| 2.1 | Flat belt selection for manufacturing catalogues, Pulleys for flat belt, V- Belt selection from catalogues, V- Grooved pulleys and Belt tension methods | | |
| 2.2 | Chain Drives: Introduction, Geometric relationship, Power rating of roller chains, | | |

	Sprockets wheels and Design of chain drives.		
3	Design of Gear Box	08 Hours	18%
3.1	Speed reducers and gear box		
3.2	Structural diagram, speed chart and Ray diagram		
3.3	Design procedure		
4	Design of Sliding Contact Bearings	07 Hours	16%
4.1	Introduction and classification of bearing		
4.2	Basics mode of lubrication, Petroff's equation, McKee's investigation		
4.3	Bearing design- selection of parameters		
4.4	Design of hydrodynamics bearing, Raimondi and Boyd's methods		
4.5	Sliding contact bearing failure- causes and remedies		
5	Design of Rolling Contact Bearings	07 Hours	16%
5.1	Introduction		
5.2	Types of rolling contact bearings, Principle of self-aligning bearing		
5.3	Static load carrying capacity, Stribeck's equation and dynamic load carrying capacity		
5.4	Load-life Relationship		
5.5	Preloading and mounting of bearings, lubrication of rolling contact bearing		
5.6	Rolling contact bearing failure- causes and remedies		
6	Design of I.C. Engine Components	08 Hours	18%
6.1	Design of cylinder, piston and piston rod		
6.2	Design of connecting rod, crank and crank shaft		
6.3	Design of Flywheel		

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, green board etc.
- Attendance is compulsory in lectures and in practical sessions.
- Internal exams / Unit tests / Surprise tests / Quizzes / Seminar / Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit /topic and will be evaluated at regular interval.
- Tutorials related to course content will be carried out in the practical.
- In the lectures discipline and behavior will be observed strictly.

- Industrial Visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of assignment.

D. Course Outcomes (COs):

On successful completion of this course, the student will be able to:

CO1: Design Gear Assembly and its elements for specific application.

CO2: Design flexible elements i.e. belt and chain drive.

CO3: Design Gear box for specific application.

CO4: Design different bearings and its components for explicit applications.

CO5: Design IC engine components for explicit applications.

CO6: Access design data book to design machine components using diverse codes as required for industrial aspects and applications.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	-	2	-	-	1	-	-	2	1	1
CO2	3	1	2	-	1	2	-	-	1	-	-	3	3	1
CO3	3	3	1	1	1	2	-	-	1	-	-	2	1	-
CO4	3	2	2	1	1	2	-	-	1	-	-	2	1	1
CO5	3	3	1	1	1	2	-	-	1	-	-	2	1	1
CO6	1	-	1	1	-	3	-	-	-	-	-	3	-	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

- Bhandari V. B., “Design of Machine Elements”, Tata McGraw Hill Publishing Co.
- Haideri Farazadak, “Machine Design “, Vol. – I, II, III, Nirali Prakashan – Pune.
- Design data book by V.B.Bhandari

Reference Books:

- Shigley Joseph Edward, “Machine Design”, McGraw–Hill Professional Publishing.
- Dieter George, “Engineering Design”, McGraw–Hill Publishing, 4th Edition 2008.
- Singh Sadhu, “Machine Design”, Khanna Publishers, New Delhi.
- George Dieter, “Engineering Design”, McGraw-Hill Publishing, 4th Edition 2008.

5. Robert L. Norton “Design of Machinery: An Introduction to the Synthesis and Analysis of Mechanisms and Machines”, Tata McGraw Hill Publishing Co.

Other material:

1. Engineering Failure Analysis, Science Direct Journal, ISSN: 1350–6307.
2. Journal of Materials and Design, ELSEVIER PUBLICATION.
3. IE Mechanical Engineering.

ME 447: POWER PLANT ENGINEERING

7th Semester and 4th Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Steam Power Plant	20
2.	Gas Turbine Power Plant	10
3.	Nuclear Power Plant	05
4.	Combined Cycle Plant	04
5.	Economics of power generation	02
6.	Jet propulsion plant	04

Total Hours (Theory): 45

Total Hours: 45

B. Detailed Syllabus:

- | | | | |
|-----|--|----------|------|
| 1 | Steam Power Plant | 20 Hours | 40 % |
| 1.1 | Classification of steam power plant. Layout of modern steam power plant. | | |
| 1.2 | Fuel handling | | |
| 1.3 | Combustion equipment for steam boilers, ash handling, dust collector | | |
| 1.4 | Chimney draught | | |
| 1.5 | Performance of Boilers, Steam condensers, Cooling ponds and cooling towers, | | |
| 1.6 | Pollution and its control. | | |
| 2 | Gas Turbine Power Plant | 10 Hours | 25% |
| 2.1 | Introduction, classification, simple open cycle gas turbine, closed cycle gas turbine, Actual Brayton cycle. | | |
| 2.2 | Means of improving the efficiency and specific output of simple cycle, open cycle gas turbine with | | |

regeneration, reheating inter cooling.

- 2.3 Effect of various modifications, effect of operating variables on air rate, actual cycle gas turbine with intercooling, reheat and regeneration, Effect of operating variables on work ratio.
- 2.4 Water injection, Combustion chambers, closed cycle gas turbine, Turbine Blade material.
- 3 Nuclear Power Plant 05 Hours 12%**
- 3.1 Introduction, Phenomenon such as fission and fusion, chain reaction, nuclear fuels.
- 3.2 General components of Nuclear reactor ,Different types of reactors, construction and working of PWR, BWR,CANDU reactor
- 4 Combined Cycle Plant 04 Hours 10%**
- 4.1 Working on gas and steam turbine – combined cycle. Arrangement combines cycle.
- 4.2 Advantages of combines cycle, parameters affecting the efficiency of combined cycle. Performance of combined cycle.
- 5 Economics of power generation 02 Hours 05%**
- 5.1 Introduction, terms and definitions, cost analysis, selection of power plant equipments.
- 5.2 Economics in plant selection
- 6 Jet and Rocket propulsion plant 04 Hours 06%**
- 6.1 Introduction to jet propulsion plant, various types of jet propulsion plants and its working.
- 6.2 Rocket propulsion – introduction and working.

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behaviour will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Explain working of different components used in steam power plant.

CO2: Evaluate performance of Boiler and Condenser.

CO3: Describe working of different nuclear power plant.

CO4: Apply energy conservation equation and heat balance and other fundamental equations of thermodynamics for gas turbine plant.

CO5: Explain working of combined cycle plant.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	1	-	-	-	-	-	-	-	-	-	2	-
CO3	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	2	1	-	-	-	-	-	-	-	-	-	1	-
CO5	2	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Power Plant Engineering by P.K.Nag, TMH Pub.
2. Power plant Engg. R.K.Rajput., Laxmi Prakashan
3. Thermal Engineering by Domkundwar and Arora., Dhanpatrai Pub

Reference Books:

1. Power Plant Engineering by Domkundwar and Arora., Dhanpatrai Pub
2. Power Plant Engineering by P. C. Sharma., S.K.Katariya and Sons
3. Power Plant Technology by G. D. Rai., Khanna Pub.
4. Power Plant Technology by El-Wakil, TMH Pub.

Other Materials:

1. www.sciencedirect.com:
International Journal of Thermal Sciences.
Experimental Thermal and Fluid Science.

2. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>
3. IEEE: www.ieeexplore.ieee.org

ME – 448 COMPUTER AIDED MANUFACTURING

7th Semester and 4th Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to CAM	02
2.	NC/CNC Machine Tools	18
3.	Flexible Manufacturing System	08
4.	Group Technology	08
5.	Rapid Prototyping	02
6.	Computer Integrated Manufacturing	07

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | | |
|-----|---|----------|-----|
| 1 | Introduction to CAM | 02 Hours | 05% |
| 1.1 | Computer Aided Design, Computer Aided Manufacturing, Reasons for CAD/CAM development, CAD/CAM system, activities, elements and applications | | |
| 2 | NC/CNC Machine Tools | 18 Hours | 40% |
| 2.1 | Fundamentals of NC/CNC Machine Tools: NC, DNC, CNC, Programmable Automation, Components of NC/CNC system, Specification of CNC system, Classification of NC/CNC Machine tools, Nomenclature of NC machine axes, CNC Control System, Automatic tool changer, Automatic Pallet Changer, Machine tool structure, Guideways, Transmission system, Drives and Feedback Devices, NC/CNC tooling | | |

2.2 Basics of CNC Programming: Coding Systems, Types of Codes, Types of Programming, Programming Functions, Basics of Turning Center and Machining Center Programming, Advanced programming and CAD/CAM based programming

3 Flexible Manufacturing System 08 Hours 17%

3.1 Introduction, objectives of an ideal FMS, applications, classification, functional components, hardware components

3.2 FMC, elements of an FMS - NC/CNC machines, coordinate measuring machines, robots, conveyors, AGVs, ASRS

3.3 FMS layouts, specifications, benefits, limitations, FMS planning and implementation issues.

4 Group Technology 08 Hours 17%

4.1 Objectives, part families, similarities, design and Manufacturing attributes

4.2 Classification methods- visual inspection, product flow analysis and coding,

4.3 Need and types of structure, coding systems, G.T. machine cells and types, concept of composite part, benefits and limitations

5 Rapid Prototyping 02 Hours 05%

5.1 Fundamentals of Rapid Prototyping, Advantages and Applications of RP

5.2 Types of Rapid Prototyping Systems

6 Computer Integrated Manufacturing 07 Hours 16%

6.1 Basic information of CIMS, hardware and software requirement for CIMS, benefits, scope and needs

6.2 CIMS wheel, elements of CIMS and their role

6.3 Fundamentals of communication, data base management

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for

the concepts being taught in lectures.

- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behaviour will be observed strictly.

D. Course Outcomes (COs):

On successful completion of this course, the student will be able to:

CO1: Understand the changes brought in the product cycles with the advent of CAM systems.

CO2: Understand emerging trends in CNC and automation.

CO3: Apply their knowledge to prepare part program for machining processes.

CO4: Understand advance philosophy in the field of manufacturing.

CO5: Outline the working behind readily available computer aided manufacturing (CAM) software.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	3	2	-
CO2	3	-	-	-	-	-	-	-	-	-	-	3	3	-
CO3	-	3	3	3	3	1	-	1	2	-	-	3	3	-
CO4	3	2	2	2		1	-	1	-	-	-	3	3	-
CO5	-	2	2	2	3	1	-	-	-	-	-	3	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Radhakrishnan P. and Subramaniam S., “CAD, CAM and CIM”, New Age International
2. Kundra T. K., Rao P. N. and Tewari N. K., “Numerical control and computer aided manufacturing”, Tata McGraw Hill Publishing company Ltd.
3. Zeid I., “CAD/CAM” , Tata McGraw Hill Publishing company Ltd.
4. Agrawal P M and Patel V J, “CNC Fundamentals and Programming”, Charotar Pub
5. Pabla and Adithan, “A Text Book of CNC Machines”, New Age International

Reference Books:

1. Kant V.S.,” Computer Integrated Manufacturing”, Prentice Hall India Pub. Company Ltd.

2. Rao P. N., "CAD/CAM", Tata McGraw Hill Publishing company Ltd.
3. Radhakrishnan P, "Computer numerical control machines", New Central Book Agency
4. Koren Y. and Joseph B.U., "Numerical Control of Machine Tools", Khanna Publishers, Delhi.
5. Sadhu Singh, "Computer Aided Design and Manufacturing", Khanna Publishers, Delhi.
6. M.P.Groover, E.W.Zimmers, "CAD/CAM Computer Aided Design and Manufacturing", Prentice Hall of India, New Delhi.
7. Kusiak A., "Intelligent Manufacturing Systems". Prentice Hall India Pub. Company Ltd.
8. Luggen W., Flexible Manufacturing Cells and System, Prentice Hall, England Cliffs, Newjersey, 1991.
9. Pham D. T. and Dimov S. S., "Rapid manufacturing", Springer Verlag, London, 2001.
10. Wohlers T., 'Wohlers Report 2007', Wohlers Associates, USA, 2007.
11. Ghosh A., "Rapid Prototyping: A Brief Introduction", Affiliated East West
12. Cooper G., "Rapid Prototyping Technology: Selection and Application", CRC Press, 2001.
13. Kai C.C., Fai L.H., Sing L.C., "Rapid Prototyping: Principles and Applications", World Scientific, 2003.

Other Material:

1. Programming Languages: C, C++, MATLAB
2. Software's: Master CAM, Pro/Engineer
3. International Journal of Mechanical Sciences (www.sciencedirect.com)
Journal of Materials Processing Technology (www.sciencedirect.com)
4. Mechanical Engg. (Inst. of Engineers) <http://www.ieindia.org/publish/mc/mc.htm>
Production Engg. (Inst. of Engineers) <http://www.ieindia.org/publish/pr/pr.htm>
5. IEEE transactions on Manufacturing Technology
<http://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=8218>
6. IET Manufacturing Engineer <http://ieeexplore.ieee.org/servlet/opac?punumber=2189>
7. Sadhna (<http://www.ias.ac.in/sadhana/>)

ME – 452 SUMMER INTERNSHIP - II
7th Semester and 4th Year

Credits and Hours:

Teaching Scheme	Project	Total	Credit
Hours/week	-	-	3
Marks	150	150	

A. Instructional Method and Pedagogy:

- Summer internship shall be at least 90 hours during the summer vacation only.
- Department/Institute will help students to find an appropriate company/industry/organization for the summer internship.
- The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline.
- Students shall commence the internship after the approval of the department Coordinator. Summer internships in research center is also allowed.
- During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and also those of the University.
- Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.
- The student shall submit two documents to the Coordinator for the evaluation of the summer internship:
 - Summer Internship Report
 - Summer Internship Assessment Form
- Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted to the Coordinator by the first day of the new term.
- The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this

guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester.

- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

B. Format of Summer Internship Report:

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- DESCRIPTION OF THE COMPANY/INDUSTRY/ORGANISATION: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
 - Location and spread of the company
 - Number of employees, engineers, technicians, administrators in the company
 - Divisions of the company
 - Your group and division
 - Administrative tree (if available)
 - Main functions of the company
 - Customer profile and market share
- INTRODUCTION: In this section, give the purpose of the summer internship, reasons for

choosing the location and company, and general information regarding the nature of work you carried out.

- **PROBLEM STATEMENT:** What is the problem you are solving, and what are the reasons and causes of this problem.
- **SOLUTION:** In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- **CONCLUSIONS:** In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- **REFERENCES:** List any source you have used in the document including books, articles and web sites in a consistent format.
- **APPENDICES:** If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.

C. Students Learning Outcomes:

On successful completion of this course, the student will be able to:

CO1: Integrate class room theory with real-world engineering problems of industry/organization

CO2: Develop work habits, attitudes and ethics necessary for industrial job.

CO3: Develop communication, interpersonal and other critical skills which can help them during job.

CO4: Develop professional report writing skill and presentation of their work in industry as well as at institute.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	-	1	-	-	-	-	-	3	-
CO2	-	-	-	-	-	3	-	3	3	2	-	-	-	-
CO3	-	-	-	-	-	2	-	-	1	3	2	2	-	-
CO4	-	-	-	-	1	-	-	-	1	3	2	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

ME492 HEATING, VENTILATION & AIR CONDITIONING

7th Semester and 4th Year

Program Elective- III

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Compound Compression	10
2.	Infiltration & Indoor Air Quality	5
3.	Cooling Load Calculation	10
4.	Duct Design, Air Distribution	10
5.	Air Conditioning applications	9

Total Hours (Theory): 45

Total Hours: 45

B. Detailed Syllabus:

- | | | | |
|-----|---|-----------------|------------|
| 1 | Compound Compression | 10 Hours | 23% |
| 1.1 | Multi stage compression system, Compound compression with intercooler, System with flash chamber, Two stage compression with flat inter cooler. | | |
| 1.2 | Multiple evaporator system, Multiple compression system, Basics of component selection. | | |
| 2 | Central air conditioning system | 08 Hours | 18% |
| 2.1 | Various types of central air conditioning system, VRF system, various types of chillers, Applied psychrometry | | |
| 2.2 | Infiltration due to stack effect, temperature difference and wind velocity, Air change and crack | | |

length methods for estimating infiltration, Infiltration due to door openings

- 2.3 Sources of indoor air pollution, Methods of control of IAQ, Fresh air requirements for ventilation and IAQ

3 Cooling Load Calculation 08 Hours 18%

- 3.1 Heat flow due to heat conduction, Solar heat load, Load from occupants, Equipment load, Infiltration air load, Miscellaneous heat sources, Fresh air load, Case studies on cooling load calculation.

- 3.2 Design of Air conditioning system, By-pass factor consideration, Effective sensible heat factor.

4 Duct Design & Air Distribution 10 Hours 23%

- 4.1 Air duct, Duct design, Pressure loss, Friction loss, Use of friction chart, Dynamic losses, Losses due to area changes, Methods of duct design, Duct material, Duct work, Duct lay out consideration.

- 4.2 Requirements of good distribution system, Terms used in air distribution, Types of Outlets, Outlet location, Selection of Air outlets, Selection procedure with the help of Nomograph, Selection procedure with tables.

5 Air Conditioning applications 09 Hours 18%

- 5.1 Ice manufacturing, Food preservation, Transport air conditioning

- 5.2 Air Conditioning in cinema halls and restaurants, District cooling plant

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Describe design aspects of various refrigeration systems and its components

CO2: Evaluate refrigeration systems to improve the performance

CO3: Recognize various types of chillers and its applications.

CO4: Apply fundamentals of cooling load calculation for any practical situation.

CO5: Solve problems related to duct design and distribution systems.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	2	-	-	2	2	-	-	-	-
CO2	3	2	2	1	1	-	-	-	-	-	-	2	2	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	3	2	1	1	-	1	-	-	-	-	2	2	-
CO5	3	2	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. “Refrigeration & Air Conditioning for engineers”, P.S. Desai, Khanna Publishers.
2. Arora. C.P., “Refrigeration & Air Conditioning”, McGraw-Hill.
3. ASHRAE Handbook

Reference Books:

1. Manohar Prasad, “Refrigeration & Air Conditioning”, New Age Publishers.
2. Rajput R. K., “Refrigeration and Conditioning”, S. K. Kataria and sons publication.
3. Domkundwar and Arora, “Refrigeration and Air Conditioning”, Dhanpat Rai & Co.
4. Jordan and Priester, “Refrigeration and Conditioning”, Prentice Hall of India.
5. Stoecker W. F., “Refrigeration and Conditioning”, Tata McGraw Hill Publication.

Web Material:

1. <http://www.nptelvideos.in/2012/12/refrigeration-and-airconditioning.html>

ME – 485 ADVANCED MATERIALS

7th Semester and 4th Year

Programme Elective-III

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	0	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Structure of Materials	08
2.	Composites Materials	11
3.	Advanced Alloys	10
4.	Nano Materials & Its Characterization	16

Total Hours (Theory): 45

Total Hours: 45

B. Detailed Syllabus:

1	Structure of Materials	08	17%
1.1	The Four Electron Quantum Numbers, Nomenclature for the Electronic States, Bonding and Energy Levels, Types of Bonds.		
1.2	Structure of Metals and Alloys, Crystal Structures and Systems, Interplanar Spacings, Types of Defects, Dislocations.		
1.3	Structure of Ceramics and Glasses, Pauling's Rules, Ceramic Crystal Structures, The Structure of Glasses.		
2	Composites Materials	11 Hours	25%
2.1	Fundamentals of composites - need for composites - Enhancement of properties - classification of composites - Matrix-Polymer matrix composites (PMC), Metal matrix composites (MMC), Ceramic matrix composites (CMC) - Reinforcement - Particle		

reinforced composites, Fiber reinforced composites. Fabrication methods of composite materials. Impact of composites on society.

3	Advanced Alloys	10 Hours	22%
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- 3.1 Alloy steels, Maraging steels, High-strength low-alloy (HSLA) steels, Dual-phase (DP) steels, Mechanically alloyed (MA) steels.
- 3.2 Superalloys, Basic alloying features, Nickel-based superalloy, Dispersion-hardened superalloys.
- 3.3 Titanium alloys, Basic alloying and heat-treatment features, types of titanium alloys,
- 3.4 Structural intermetallic compounds, General properties of intermetallic compounds, Nickel aluminides, Titanium aluminides, Other intermetallic compounds, Aluminum-lithium alloys,

4	Nano Materials & Its Characterization	16 Hours	36%
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- 4.1 Introduction to advanced materials, Current applications of advanced nanostructured Materials/thin film.
- 4.2 Thin film coatings & its techniques. Introduction to Physical Vapor Deposition (PVD) processes, various types of PVD processes.
- 4.3 Introduction to characterization of coatings, X-ray diffraction (XRD), Scanning Electron Microscopy (SEM), Atomic Force Microscopy (AFM), Transmission Electron Microscopy (TEM).

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of this course, the student will be able to:

CO1: Understand the structure of advanced materials.

CO2: Identify the types of composite materials.

CO3: Development of composite materials with various fabrication methods.

CO4: Know about advanced alloys and its applications.

CO5: Characterization of different coatings surfaces.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	1	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	2
CO4	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	2	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Raghavan V., “Materials Science and Engineering”, Prentice-Hall of India Private Limited 2003.
2. Smith W. F., “Principles of Materials Science and Engineering”, McGraw Hill, New York 1994.
3. Reid Hill R. E., “Physical Metallurgy Principles”- PWS-Kent Publishing 2004.
4. Callister W. D., “An Introduction Materials Science & Engineering”, John Wiley & Sons 2007.
5. Mathews F.L. and Rawlings R.D., Composite materials: Engineering and Science, Chapman and Hall, London, England, 1st edition, 1994.
6. Chawla K.K., Composite materials, Springer Verlag, 1987.
7. P.M. Martin, "Handbook of Deposition Technologies for Films and Coatings: Science, Applications and Technology", Elsevier USA (2010).

Reference Books:

1. Van Vlack L.H., “Elements of Materials Science and Engineering”, Addison Wisley, New York.
2. Smallman R. E. and Ngan A. H.W., “Physical Metallurgy and Advanced Materials”, Elsevier, UK
3. Mitchell B. S., “An introduction to Materials Engineering and Science”, John Wiley & Sons, New Jersey, 2004

Other Materials:

1. Journal of Material Science and Technology.
2. Journal of Material Chemistry.
3. Advanced Materials Journal.
4. Nanotechnology Journal.
5. Advanced Functional Materials Journal.
6. http://nptel.iitm.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/Manuf%20Proc%20II/New_index1.html
7. <http://www.sciencedirect.com/science/journal>
8. <http://www.ieindia.org/publish/pr/pr.htm>
9. <http://www.ias.ac.in/sadhana>

ME484 DESIGN OF HEAT EXCHANGERS

7th Semester and 4th Year

Programme Elective-III

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	–	3	3
Marks	100	–	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to heat exchangers and its classification	06
2.	Design of double pipe heat exchangers	12
3.	Design of shell & tube heat exchangers	13
4.	Compact & phase change heat exchangers	10
5.	Design of cryogenic heat exchanger	04

Total Hours (Theory): 45

Total Hours: 45

B. Detailed Syllabus:

- 1 Introduction to heat exchangers and its classification 06 Hours 12%
Basic design methodologies: Classification of heat exchanger, selection of heat exchanger, Thermal-Hydraulic fundamentals, Overall heat transfer coefficient, LMTD method for heat exchanger analysis for parallel, counter, multi-pass and cross flow heat exchanger, e-NTU method for heat exchanger analysis, Fouling, Rating and sizing problems, heat exchanger design methodology
- 2 Design of double pipe heat exchangers 12 Hours 30%
 - 2.1 Thermal and Hydraulic design of inner tube and annulus, hairpin heat exchanger with bare and finned inner tube, total pressure drop

3	Design of shell & tube heat exchangers	13 Hours	30%
3.1	Basic components, basic design procedure of heat exchanger, TEMA code, J-factors, conventional designs methods, Bell-Delaware method.		
4	Compact & phase change heat exchangers	10 Hours	20%
4.1	Heat transfer enhancement, plate fin heat exchanger, tube fin heat exchanger, heat transfer and pressure drop		
4.2	Condensers classification and design methods for surface condensers, Evaporators Classification and design methods		
5	Design of cryogenic heat exchanger	04 Hours	8%
5.1	Design of heat exchangers for liquefaction systems ,single tube ,double tube Linde heat exchangers three channel heat exchangers ,multiple tube type ,Giauque Hampton and Collins type heat exchangers		
5.2	Finned tube and plate type heat exchangers, different configuration heat transfer coefficients and friction coefficient for various configurations.		

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behaviour will be observed strictly.

D. Students Learning Outcomes:

On successful completion of the course, the student will be able to:

CO1: Evaluate performance parameters and rate different types of heat exchangers.

CO2: Design double pipe heat exchangers.

CO3: Design compact and shell & tube type heat exchangers.

CO4: Design various phase change and cryogenic heat exchangers.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	1	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	3	1	1	-	-	-	-	-	-	-	1	-
CO3	3	2	3	1	1	-	-	-	-	-	-	-	1	-
CO4	2	1	1	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Ramesh K Shah, “Fundamentals of Heat Exchanger Design”, Wiley Publication. 2015.
2. Heat Exchangers by Sadik Kakac, Hongtan Liu, Anchasa Pramuanjaroenkij, CRC Press
3. Compact Heat Exchangers, Kays & London,

Reference Books:

1. Kuppan, T, Macel Dekker, “Heat Exchanger Design Handbook”, CRC Press, 2017.
2. Schunder E.U., “Heat Exchanger Design Hand Book”, Hemisphere Pub.1983.
3. Joshi, M.V., & Mahajani, V.V., Process Equipment Design, Macmillan, India, Ltd., 1996.
4. Donald Q Kern, “Process Heat transfer”, Tata McGraw Hill, 2017

Web Material:

1. <https://nptel.ac.in/courses/112/105/112105248/>

ME483 INDUSTRIAL AUTOMATION & ROBOTICS

7th Semester and 4th Year

Program Elective- III

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	-	3	3
Marks	100	-	100	

A Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Automation	03
2.	Automated Production system and assembly system	05
3.	Machine Vision	05
4.	Fundamental of Industrial Robotics	02
5.	Robotic End Effectors, Control System and Sensors	05
6.	Robot Programming	03
7.	Robot Arm Kinematics and Dynamics	16
8.	Trajectory Planning	06

Total hours (Theory): 45

Total hours: 45

B Detailed Syllabus:

- | | | | |
|-----|---|----------|-----|
| 1 | Introduction to automation | 03 Hours | 07% |
| 1.1 | Introduction, applications, Elements of automation, Goal of automation | | |
| 1.2 | Advantages and disadvantages of automation, Types of Automation, Low cost automation | | |
| 2 | Automated Production system and assembly system | 05 Hours | 11% |
| 2.1 | Introduction, Transportation device and feeders. | | |
| 2.2 | Application of automated production lines automated assembly system, Analysis of transfer lines | | |

3	Machine vision	05 Hours	11%
3.1	Introduction and applications		
3.2	Components of a machine vision system		
3.3	Image acquisition and representation		
4	Fundamental of Industrial Robotics	02 Hours	05%
4.1	Specifications, Characteristics, Components, Configurations, Selection criteria and Applications		
5	Robotic end effectors, control system and sensors	05 Hours	11%
5.1	Robot End effector: Introduction, Types and classification , Design of gripper, RCC		
5.2	Robot Drives and control: Hydraulic, Pneumatic and electrical Actuators and controls		
5.3	Robot transducer and sensors		
6	Robot programming	03 Hours	07%
6.1	Methods of programming		
6.2	Study of various programming languages		
7	Robot arm kinematics and dynamics	16 Hours	35%
7.1	Direct kinematics and solutions		
7.2	Inverse kinematics and solutions		
7.3	Robot arm dynamics, lagrange and newton euler formulations		
7.4	Generalised D'Alembert's equation of motion		
8	Trajectory Planning	06 Hours	13%
8.1	Introduction, basic trajectory Planning		
8.2	Joint-Space Trajectory Planning, Cartesian-Space Trajectories, Continuous Trajectory Recording		

C Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.
- Students will visit various industries/research centers/university departments in the vicinity to observe the manufacturing setups/software.

D Course Outcomes (COs):

At the end of the course, the students will be able to

CO1: Explain different automation methods used in assembly and transfer line.

CO2: Interpret the different components of machine vision and explain different type of sensor used in robotics system.

CO3: Explain different configuration and type of industrial robot.

CO4: Design and analyses the effector or gripper used in robotics.

CO5: Evaluate robotic motion by solving kinematics and dynamics of robot arm manipulator with theory and some software.

CO6: Plan the trajectory and prepare the program for motion of robot arm.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	-	-	2	1	1	-	-	-	-	2	1
CO2	2	2	2	1	-	-	1	-	2	2	-	-	2	-
CO3	2	1	-	-	-	1	-	1	-	-	-	-	2	-
CO4	3	3	3	2	-	-	2	-	2	2	-	-	2	-
CO5	3	3	3	3	3	-	-	-	2	2	-	1	2	-
CO6	2	2	2	2	-	-	-	-	-	-	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E Recommended Study Material:

Text Books/ Reference Book:

1. Groover M P, “Automation, Production Systems and Computer Integrated Manufacturing”, Third Edition Pearson Educational Publication

2. Boothroyd Geoffrey, "Assembly Automation and product design", Second Edition, CRC Press
3. Khaimoswitch E M, "Hydraulic control of machine tools", Pergamon press
4. Gupta A K and Arora S K, "Industrial Automation and Robotics", Third Edition, Laxmi Publication Ltd.
5. Klafter R D, Chmielewski T A and Negin M, "Robot Engineering: An Integrated approach", PHI
6. Shahinpoor Moshen, "A Robot Engg Text Book", Harper and Row Publishers, NY.
7. Groover and Zimmer, "CAD/CAM: computer-aided design and manufacturing", Fifth edition Prentice-Hall.
8. Thomas H A, "Handbook of low cost automation techniques", Gower, 1969
9. Groover M P, "Industrial Robotics Technology, programming and applications", McGraw-Hill
10. Schilling Robert J, "Fundamentals of Robotics, Analysis and Control", Prentice Hall of India
11. Craig John J, "Introduction to Robotics, Mechanics and control", 3rd Ed, Addison – Wesley
12. David Vernon, "Machine Vision", Prentice Hall, New York.
13. Mittal R K and Nagrath I J, "Robotics and Control", McGraw-Hill, 2013.
14. Saha S K, "Introduction to Robotics", McGraw-Hill, 2015, Second edition.

Reading Materials, web materials with full citations:

1. <http://nptel.iitm.ac.in>
2. Simbad 3D, Robo Works 3.0, Lab View 2010
3. Production Engg. (Inst. of Engineers)- <http://www.ieindia.org/publish/pr/pr.htm>
4. Journal of Mechanisms and Robotics (ASME)- <http://asmedl.aip.org/JMR>
5. Journal of Robotics and Automation (IEEE)
<http://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=56>
6. Sadhna (<http://www.ias.ac.in/sadhana/>)
7. Mechanical Engg. (Inst. of Engineers) <http://www.ieindia.org/publish/mc/mc.htm>
8. IEEE Transactions on Automation Science and Engineering
(<http://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=8856>)
9. IET Control and Automation
(<http://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=4469873>)

ME475 PRODUCT DESIGN & VALUE ENGINEERING

7th Semester and 4th Year

Program Elective- III

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	03	00	03	03
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Product Design Fundamentals.	04
2.	Product design ASIMO's model	10
3.	Role of Allowance, Process Capability, and Tolerance in Detailed Design and Assembly	04
4.	Sustainable product design	05
5.	Value Management	06
6.	Value engineering Methodology	08
7.	VE Success Stories	08

Total Hours (Theory): 45

Total Hours: 45

B. Detailed Syllabus:

1	Product Design Fundamentals	04 Hours	09%
1.1	The Design Process.		
1.2	The Product concept.		
1.3	The Product Research		
1.4	Essentials of product design.		

- 1.5 Product development criteria: Functionality, Reliability, Durability, Serviceability, Aesthetics, Ergonomics, Design for Manufacture and Design for Assembly Standardization, Simplifications
- 1.6 Product Development Tools: (Brief overview): CAD, QFD, FMEA, FEA, Concurrent Engg.
-
- 2 Product design ASIMO's model 10 Hours 22 %**
- 2.1 Phase I to phase VII
- 2.2 Primary Design Phases and Flowcharting
- 2.3 Product Development Challenges: Creativity, Cost, time, team management, customer segmentation, ever changing needs of customers.
- 2.4 Product Development Plan: Selection of product, evaluation of alternatives, resource allocation, customer needs, product specifications
-
- 3 Role of Allowance, Process Capability, and Tolerance in Detailed Design and Assembly 04 Hours 09%**
- 3.1 Allowance
- 3.2 Process Capability
- 3.3 Tolerance in detailed design and assembly
-
- 4 Sustainable product design 05 Hours 10%**
- 4.1 History of sustainable design
- 4.2 Material and processes.
- 4.3 Pleasant experiences of sustainable design
-
- 5 Value Management 06 Hours 14%**
- 5.1 Value engineering concepts
- 5.2 Evaluation of Ideas
- 5.3 Value Engineering and Value Analysis, Objectives of Value Analysis, When to apply Value Analysis
-
- 6 Value engineering Methodology 08Hours 18%**
- 6.1 Anatomy of the function
- 6.2 Application of value engineering phases.
- 6.3 Value engineering techniques.

7 VE Success Stories

08 Hours 18%

7.1 Case studies associated with value analysis and value engineering.

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lecture's behaviour will be observed strictly.

D. Students Learning Outcomes:

On successful completion of the course, the student will be able to:

CO1: Identify product design and development steps.

CO2: Explain the different types of sustainable design options for product.

CO3: Identify the functions of product or service.

CO4: Explain value Engineering and its techniques.

CO5: Solve the product scheduling problems working.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	2	2	-	-	-	-	-	-
CO3	-	-	-	2	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO5	-	-	-	-	4	-	-	-	-	2	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Kumar Prashant, “Product Design: Creativity, Concepts and Usability”, PHI Learning Pvt. Ltd. 2012, 2012.,

2. Anil Kumar Mukhopadhyaya Value Engineering Mastermind: From Concept to Value Engineering Certification SAGE Publications India, 2009.
3. R C Gupta A K Chitale Product design and manufacturing Phi Learning Pvt. Ltd., New Delhi 6TH Edition, 2013
4. Dr. M. A. Bulsara and Dr. H. R. Thakkar product design and value engineering Charotar Publishing House Pvt. Ltd 2nd Edition : 2015

Reference Books:

1. Richard Morris, “The Fundamentals of Product Design”, Bloomsbury Publishing NewYork, 2nd Edition. 2017,
2. Jonathan Chapman, “Routledge Handbook of Sustainable Product Design”, Routledge, 2017
3. Karl T. Ulrich, “Product Design and Development”, Tata McGraw-Hill Education, 2003
4. Devdas Shetty, Product Design For Engineers “Cengage Learning, 2015
5. Mudge, Arther E. 1971, Value Engineering: A Systematic Approach, McGraw-hill: New York.
6. Alphonse Dell’Isola, “Value Engineering: Practical Applications for Design, Construction, Maintenance & Operations”, R S Means Co., 1997
7. Richard Park, “Value Engineering: A Plan for Invention”, St. Lucie Press, 1999.
8. Del L. Younker, “Value Engineering analysis and methodology”, Marcel Dekker Inc, New York, 2004.
9. Mike Baxter., “Product Design”, CRC Press, 1995

Web Material:

1. NPTEL COURSE .<http://nptel.ac.in/courses/112104230/12>

Other Material:

1. <https://www.value-eng.org>.
2. [https:// www.invest-in.org](https://www.invest-in.org).
3. <https://www.sciencedirect.com>(Journal of Product Innovation Management)
4. <http://www.inderscience.com> (International Journal of Product Development)

ME487 PROJECT MANAGEMENT

7th Semester and 4th Year

Program Elective- III

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	00	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Fundamentals of Project Management	05
2.	Planning the Project	06
3.	Budgeting the project	06
4.	Scheduling the project	06
5.	Allocating resources to the project.	06
6	Monitoring and controlling the project.	06
7	Evaluating and Terminating the project	04
8	Project Management Related Quality standards and International projects features.	06

Total Hours (Theory): 45

Total Hours: 45

B. Detailed Syllabus:

1	Fundamentals of Project Management	05 Hours	10%
1.1	What is project		
1.2	Project management vs General Management.		
1.3	Project managements role and responsibilities		

1.4	Matrix Project organization..		
2	Planning the Project	06 Hours	14%
2.1	Planning process.		
2.2	Work breakdown structure and other aids		
2.3	Multidisciplinary team management.		
3	Budgeting the project	06Hours	13%
3.1	Methods of budgeting project.		
3.2	Cost Estimating and improvements.		
3.3	Budget Uncertainty and Risk Management.		
4	Scheduling the project	06 Hours	13%
4.1	Project uncertainty.		
4.2	Simulation analysis for project completion times		
4.3	The Gantt Chart.		
4.4	Extensions of PERT and CPM.		
5	Allocating resources to the project.	06 Hours	13%
5.1	Expediting a project by using Excel.		
5.2	Resource loading.		
5.3	Allocating scarce resources to projects.		
5.4	Goldratt's Critical Chain.		
6	Monitoring and controlling the project	06 Hours	13%
6.1	The Plan –Monitor-Control-Cycle		
6.2	Data collecting and Reporting.		
7	Evaluating and Terminating the project	04 Hours	10%
7.1	Project Evaluation criteria and measurement.		
7.2	Project auditing.		
7.3	Project Termination		
8	Project Management -Quality standards & International projects features	06 Hours	14%

8.1 Quality standards

8.2 Speculative features of International projects.

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Recognize project management concept. and use of work breakdown structures (WBS) in a project application

CO2: Explain the cost estimation procedure in project and Describe the risk management in a project.

CO3: Explain budgeting methods. and Know how to collect data collection and reporting in a project

CO4: Identify evaluation criteria and discuss major features of International project Management

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	-	-	2	1	1	-	-	-	-	2	1
CO2	2	2	2	1	-	-	1	-	2	2	-	-	2	-
CO3	2	1	-	-	-	1	-	1	-	-	-	-	2	-
CO4	3	3	3	2	-	-	2	-	2	2	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Chaudhary S., "Project Management ", TMH ;, 3rd Edition.2004
2. Chandra Prasanna, "Projects: Planning, Analysis, Selection, Financing, Implementation and Review", Tata McGraw Hill, 7th Edition, New Delhi, 2008.
3. Gupta Rajeev, "Project Management", PHI learning solutions, 2nd Ed. New Delhi, 2014
4. Greig Horine ., " Project Management Absolute Beginner's Guide; 3rd edition QUE ,2012
5. Wouter Baars Project Management Handbook 1st edition 2006

Reference Books:

1. Samuel J Mantel, "Project Management", Wiley India Edition, 2006
2. Joseph Phillips, Project Management fpr small business, Harper Collins Pub., USA 2012
3. Project Management Institute, " Project Management Institute; 6th ed edition 2017

Other Material:

1. <https://www.edx.org/course/introduction-to-project-management>
2. [https://www.edx.org/professional-certificate/delftx-management-of-engineering-projects-dealing-](https://www.edx.org/professional-certificate/delftx-management-of-engineering-projects-dealing-with-complexity)
3. [with-complexity -](https://www.edx.org/professional-certificate/delftx-management-of-engineering-projects-dealing-with-complexity)
4. <https://www.edx.org/course/international-project-management>
5. <https://lecturenotes.in/subject/171/project-management-PM>
6. <https://projektkvalitet.dk/wp-content/uploads/the-practical-guide-to-project-management.pdf>
7. <https://www.manage.gov.in/studymaterial/PM.pdf>
8. <https://examupdates.in/project-management-notes/>

ME488 MECHANICAL ESTIMATION & COSTING

7th Semester and 4th Year

Program Elective- III

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	00	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Fundamentals of estimating and costing	05
2.	Material cost Estimation.	05
3.	Machine shop- cost estimation	07
4.	Sheet Metal cost -estimation	07
5.	Forging shop- estimation	07
6.	Welding shop -estimation	07
7.	Foundry shop estimation including pattern making	07

Total Hours (Theory): 45

Total Hours: 45

B. Detailed Syllabus:

1	Introduction to Estimating and costing	05 Hours	10%
1.1	Comprehensive Meaning of Estimating		
1.2	Meaning of costing		
1.3	Costing elements and calculations		
1.4	Requirement of product estimation as per customer needs		
1.5	Role of various softwares pertaining to Mechanical estimating and costing		

2	Material cost Estimation	05 Hours	10%
2.1	Procedure of calculating material cost		
2.2	Calculations for different shapes of material		
3	Machine shop Estimation	07 Hours	16%
3.1	Time considerations in Machine shop		
3.2	Technical terms pertaining to Machine shop estimation.		
3.3	Estimations in different operations such as Turning drilling, Milling, shaping ,planning etc.		
4	Sheet Metal cost -estimation	07 Hours	16%
4.1	Types of Sheet materials required in sheet metal shop		
4.2	Sheet thickness and gauges and its use		
4.3	Operations in sheet metal shop		
4.4	Calculations in sheet metal along with some actual examples		
5	Forging shop cost estimation.	07 Hours	16%
5.1	Procedure forging including different losses		
5.4	Materials Used in forging along with use and specifications		
6	Welding shop -estimations	07 Hours	16%
6.1	Main components of welding cost in different welding's		
6.2	Factors affecting welding costs		
6.3	Total weld cost analysis		
7	Foundry shop estimation	07 Hours	16%
7.1	Different elements of cost in foundry shop		
7.2	Different elements of cost in Pattern Making shop		
7.3	Costing for selling price of casting products		

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.

- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Identify the meaning and need of Estimating and costing.

CO2: Explain the Procedure for material cost estimation

CO3: Explain the different types of sheets and related estimations in sheet metal working.

CO4: Calculate cost estimation in different forging operations and identify Different elements in estimation in foundry shop

CO5: Calculate components of welding estimations.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	2	-	-	-	-	-	-	-	-	-	-	1	-
CO3	-	2	-	-	-	-	-	-	-	-	-	-	-	-
CO4	2	2	-	-	-	-	-	-	-	-	-	-	-	-
CO5	1	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Banga T.R., “Mechanical Estimating and costing”, Khanna Publishers,, 16th Edition.2003
2. B. P. Sinha., “Mechanical Estimating and Costing”, McGraw-Hill Education, New Delhi ,1995
3. M. Adithan., “Process Planning and Cost Estimation”, New Age International (P) Limited, New Delhi 2nd Edition.2003,2015

Reference Books:

1. Kuang-Hua Chang, “Product Manufacturing and Cost Estimating Using CAD/CAE”, Khanna Elsevier Science.,2013

2. A. K. Chitale, r. C. Gupta, “Product Design And Manufacturing”, Khanna Prentice-Hall Of India Pvt, New Delhi , 2nd Edition 2007
3. D' Amelio, Joseph, “Mechanical Estimating Manual - Sheet Metal, Piping and Plumbing”, Fairmont Press, Inc., 1st Edition, 2007.

Other Material:

1. <https://www.esabasia.com/asia/en/support/value-added-engineering/quick-weld-productivity-analyzer.cfm>
2. <https://www.fabricatingandmetalworking.com/2012/04/estimating-total->
3. <https://www.twi-global.com/technical-knowledge/job-knowledge/welding-costs-096>
4. <http://www.calculatoredge.com/matweight/material%20wt.htm>
5. <http://www.calculatoredge.com/index.htm#mechanical>
6. <https://www.easycalculation.com/engineering/civil/wood-cft.php>

ME493 FINITE ELEMENT METHODS

7th Semester and 4th Year

Program Elective- III

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	00	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Finite Element Analysis	05
2.	Finite Element Formulation Two-Dimensional Problems	25
3.	Finite Element Analysis in different Engineering applications	15

Total hours (Theory): 45

Total: 45

B. Detailed Syllabus:

1. Introduction to Finite Element Analysis 05 Hours 11%
 - 1.1 Introduction; Historical Background, Stresses and Equilibrium, Boundary Conditions.
 - 1.2 Strain-displacement Relations, Stress-strain Relations, Temperature Effects.
 - 1.3 Finite Element Formulation Techniques: Virtual Work and Variational Principle, Galerkin Method, Displacement Approach, Stiffness Matrix.
 - 1.4 Element Properties: Coordinates and a Shape Functions, Triangular Elements, Rectangular Elements, Solid Elements etc.
2. Finite Element Formulation Two-Dimensional Structural Problems 25 Hours 56%
 - 2.1 Analysis of Truss, Continuous Beam and Frame
 - 2.2 Finite Element Modeling for Axis Symmetric Problems Formulation

2.3 FEM for Plates and Shells

3 Finite Element Analysis in different Engineering applications 15 Hours 33%

3.1 Finite Element formulation for Heat Transfer Problems.

3.2 Finite Element Modeling for Potential Flow

3.3 Finite Element Analysis of Dynamic Problems

C. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures behaviour will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Apply the knowledge of FEM for 2D stress analysis, heat transfer analysis and flow analysis.

CO2: Formulate and solve problems of trusses, beams, planar loading and axisymmetric.

CO3: Formulate and solve preliminary problems for dynamic analysis.

CO4: Interpretate and evaluate the quality of results obtained using FE software.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	2	2	-	-	-	-	-	-	-	3	-
CO2	3	3	-	2	2	-	-	-	-	-	-	-	3	-
CO3	3	3	-	2	2	-	-	-	-	-	-	-	3	-
CO4	3	3	3	2	3	-	-	-	-	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Chandrupatla and Belagundu, "Finite Elements in Engineering", Prentice Hall of India Private Ltd., 1997.
2. Rao S.S., "Finite Element Method in Engineering", Elsevier Pergaman Press, 1997.
3. Logan, D. L. "A first course in the finite element method. Cengage Learning" 2016.

Reference books:

1. Seshu, P., Textbook of Finite Element Analysis, Prentice-Hall, India, 2003.
2. George R. Buchaman, "Schaum's Outline of Finite Element Analysis", McGraw Hill Company, 1994.
3. Reddy J.N., "An Introduction to the Finite Element Method", McGraw Hill, Int. Edition, 2005.
4. Cook Robert Davis, "Concepts and Applications of Finite Element Analysis", John Wiley and Sons, 1999.
5. David V. Huton, "Fundamentals Of Finite Element Analysis" Tata McGraw Hill, Edition 2005.

Web Material:

1. <http://www.mece.ualberta.ca/Tutorials/ansys/>
2. <http://www.ansys.com>
3. <http://www.owlnet.rice.edu/~mech403/FEA>

Other Material:

1. Software: ANSYS
2. Finite Elements in Analysis and Design, an International Journal for Innovations in Computational Methodology and Application, ELSEVIER Publication.

ME490 ADVANCED MACHINE DESIGN

7th Semester and 4th Year

Program Elective- III

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	00	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Design Basics and Philosophy	06
2.	Design for strength, rigidity and anisotropy	06
3.	Design based on Fatigue and Creep	07
4.	Design based on Fracture	06
5.	Advances in gear design	10
6.	Ergonomics, DFM, QFD-CE, PDD-PC	05
7.	Design of material handling equipment	05

Total hours (Theory): 45

Total: 45

B. Detailed Syllabus:

1	Design basics and philosophy	04 Hours	9%
1.1	General design procedure for design problems		
1.2	Design concepts		
1.3	Safe life v/s fail safe design		
1.4	Formation of design team		
1.5	Reliability based design of mechanical components		
1.6	Tribological aspects in machine design		
2	Design for strength, rigidity and anisotropy	06 Hours	13%
2.1	Design for strength and rigidity		

2.2	Theory of elasticity and plasticity with design analysis		
2.3	Design using anisotropic materials eg. composite materials		
3	Design based on Fatigue and creep	07 Hours	15%
3.1	Design against Fatigue		
3.2	Design against creep		
4	Design based on fracture	06 Hours	13%
4.1	Overview of fracture mechanics		
4.2	Common applications of fracture/fracture prevention		
4.3	Historical development		
4.4	LEFM and EPFM		
4.5	Modes of Loading		
4.6	SIF for various geometries and Loading		
5	Advances in gear design	10 Hours	24%
5.1	Advances in gear design		
5.2	Design of corrective gear		
5.3	Types of corrected gearing		
5.4	Correction factor to avoid undercutting		
5.5	Peaking		
5.6	Tooth thickness of corrected gears at the pitch circle		
5.7	Numerical on corrective gear design		
6	Ergonomics, DFM, QFD-CE, PDD-PC	06 Hours	13%
6.1	Ergonomics: Human factors considerations in design and applications,		
6.2	DFM: Design for manufacturing – including assembly aspects and other aspects,		
6.3	QFD-CE: Quality function deployment – concurrent engineering,		
6.4	PDD-PC: Product design and development-product cycle		
7	Design of material handling equipment	06 Hours	13%
7.1	Recent trends in materials handling equipment design		
7.2	Objectives of Material Handling System		

- 7.3 Types of loading, Bulk Material Character Classification, Containerization
- 7.4 Methods of loading and discharging of conveyors
- 7.5 Constructional details of a conveyor belt
- 7.6 Conveyor pulleys
- 7.7 Force analysis in belt conveyors
- 7.8 Amazon's Robotic Warehouses the Future of Material Handling

C. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures behaviour will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Develop the ability to make proper assumptions, perform correct analysis while designing specific mechanical components.

CO2: Describe the material handling equipment, its design principles, different loading patterns and structural analysis.

CO3: Analyse the various modes of failure of machine components under different load patterns

CO4: Use design data books and different codes of design.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	1	-	-	-	-	-	-	-	-	-	3	-
CO2	2	2	1	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	1	-	-	-	-	-	-	-	-	-	3	-
CO4	1	1	1	-	-	-	-	-	-	-	-	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Bhandari V. B., “Design of Machine Elements”, Tata McGraw Hill Publishing Co.
2. Farazdak Haideri, “Mechanical System Design”, Nirali Prakashan, 3rd Edition.

Reference books:

1. George Dieter, “Engineering Design”, McGraw-Hill Publishing, 4th Edition 2008.
2. Robert L. Mott, “Machine Elements in Mechanical Design”, Prentice Hall, 4th Edition.
3. Robert C. Juvinall and Kurt M. Marshek, “Fundamentals of Machine Component Design”, Wiley Publication.
4. Robert L. Norton “Design of Machinery: An Introduction to the Synthesis and Analysis of Mechanisms and Machines”, Tata McGraw Hill Publishing Co.
5. Burr and Cheathaam, Mechanical analysis and design.
6. G.M. Maitra, Handbook of Gear design. vol –I and II
7. W. D. Callister, Material Science.
8. Material handling equipment by P. Rudenko.

Web Material:

1. ASME Journal of Mechanical Design (<http://asmedl.aip.org/MechanicalDesign>)
2. <https://prostack.com/2018/04/12/amazons-robotic-material-handling/>

Other Materials:

1. Ramesh K, IIT Madras India, E-book on Engineering Fracture Mechanics.
2. PSG Design Data Book
3. Abdulla Shariff, Hand Book of Properties of Engineering Materials and Design Data for Machine, Elements
4. Design data book by V. B. Bhandari.

ME491 ARTIFICIAL INTELLIGENCE

7th Semester and 4th Year

Program Elective- III

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to AI	05
2.	Problem Solving Approaches	06
3.	Knowledge Representation and Reasoning	06
4.	Machine Learning	08
5.	Artificial Neural Networks	10
6.	Applications of AI in Mechanical Engineering	10

Total Hours (Theory): 45

Total Hours: 45

B. Detailed Syllabus:

1. Introduction to AI 05 hours 12%
 - 1.1 Background of AI: Impact and Examples of AI
 - 1.2 AI Concepts and Terminology-Cognitive Computing- Perception, Learning and Reasoning
 - 1.3 Scope of AI in various fields like natural language processing, vision and speech processing, robotics, expert systems

2	Problem Solving Approaches	06 Hours	13%
2.1	Problem Solving by search: Heuristic Search, Informed Search		
2.2	Constraint Satisfaction Problems,		
2.3	Depth first search, breadth-first search, branch and bound		
3	Knowledge Representation and Reasoning	06 Hours	13%
3.1	Introduction to Knowledge Representation		
3.2	Propositional Logic		
3.3	First Order Logic		
3.4	Inference in First Order Logic		
4	Machine Learning	08 Hours	18%
4.1	Introduction to Machine Learning		
4.2	Types of Machine Learning		
4.3	Linear Regression, Logistic Regression		
4.4	Decision tree		
5	Artificial Neural Networks	10 Hours	22%
5.1	Neural Networks Overview		
5.2	Neural Network Representation		
5.3	Computing a Neural Network's Output		
5.4	Activation Functions, Need for Non-Linear Activation Functions		
5.5	Gradient Descent for Neural Networks		
5.6	Backpropagation,		
5.7	Brief Introduction to Deep Learning		
5.8	ANN in Mechanical characterization		
5.9	ANN in Tribological characterization		
6	Applications of AI in Mechanical Engineering	10 Hours	22%
6.1	Applications of AI techniques in Design and Optimization of Mechanical Components (Including Materials selection, Dimensions and Analysis)		
6.2	AI in Traditional and Non- Traditional Machining Processes, Fabrications, Forming processes like Welding, Rolling, Casting, Forging etc. Process Planning and Quality		

Control

6.3 Fuzzy logic & its applications in industry

6.4 AI in industry 4.0

6.5 AI in robotics

6.6 AI in automotive

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as apart of continuous internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciationfor the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Describe how machines act and think.

CO2: Perform the theorem proving and game playing along with reasoning, learning and planning.

CO3: Apply the concepts of neural network and machine learning for the understanding the behaviour of machines.

CO4: Demonstrate ANN as a computing tool in the mechanical and tribological characterization.

CO5: Apply AI in mechanical engineering field such as design, industry 4.0, automotive, robotics, etc.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	2	-	-	-	-	-	-	-	-	-	2	-
CO2	2	2	2	-	-	-	-	-	-	-	-	-	-	-
CO3	2	2	2	-	3	-	-	-	-	-	-	-	2	-
CO4	3	3	3	3	3	-	-	-	-	-	-	-	1	2
CO5	3	3	3	3	3	-	-	-	-	-	-	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Rushell and Norvig, “Modern Approach to Artificial Intelligence”, Prentice Hall of India Ltd., 2006
2. Patterson D W, “Artificial Intelligence And Expert Systems”
3. Rich E and Knight K, “Artificial Intelligence”, (2nd Edition) Tata McGraw-Hill

Reference Books:

1. Rajasekaran S and Vijayalakshmi Pai G A, “Neural Network, Fuzzy Logic and Genetic Algorithms synthesis and Applications”, PHI Publication
2. Rajeshkaran S, “Hybrid Intelligent Systems”, PHI
3. Ian Goodfellow, Yoshua Bengio and Aaron Courville, “ Deep Learning”, MIT Press book.
4. Townsend Carl, “Introduction to Prolog Programming”
5. Bratko Ivan, “PROLOG Programming For Artificial Intelligence”, (Addison-Wesley)
6. Klocksins and Mellish, “Programming with PROLOG”.
7. D. W. Rolston, “Artificial Intelligence and Expert System, Development”, McGraw-Hill International Edition.
8. Jang J S R, Sun C T, Mizutani, “Neuro-Fuzzy and Soft Computing”, Pearson Education Edition, 1997, Reprint 2007

9. Akerker and Sajja, "Knowledge-Based Systems, Jones and Bartlett", MA, USA, 2009
10. Ian Cloete and Jacek Zurada, "Knowledge Based Neuro-Computing", University Press, Massachusetts Institute of Technology, USA, 2002
11. Oscar Cordon, Francisco Herrera, Frank Hoffmann, Luis Magdalena, "Genetic Fuzzy Systems", Word Scientific Publishing Ltd. , 2001
12. Partihar D K, "Soft Computing", Narosa Publication, 2008
13. Sivanandam S N and Deepa S N, "Principles of Soft Computing", Wiley India Edition, 2007

Web Materials:

1. <http://www.aaai.org/aitopics/html/overview.html>
2. <http://www.devarticles.com/c/a/Java/AIBased-Problem-Solving/>
3. <http://www.cse.unsw.edu.au/~billw/aidict.html#backwardchaining>
4. <http://www.cogsci.ucsd.edu/~batali/108b/lectures/heuristic.html>
5. <http://www.jfsowa.com/pubs/semnet.htm>
6. <http://www.cs.sunysb.edu/~warren/xsbbook/node3.html>
7. http://www.doc.ic.ac.uk/~nd/surprise_96/journal/vol4/cs11/report.html
8. <http://neurointelligence.com>
9. www.soft-computing.de

Other Materials:

1. Programming Languages: PROLOG
2. Software's: Alyuda Neuro Intelligence, Neuro Solutions 5.1, Easy NN Plus, Weka
3. www.ieeexplore.ieee.org
4. Computer Engineering (IE)
5. Jr. of Computing and Information science engineering
6. Jr. of Artificial Intelligence Research (AI Access Foundation)
7. Jr. of Material Processing Technology (Sciencedirect)
8. Jr. of Scientific and Industrial Research (NISCAIR, New Delhi)

ME 495: THERMAL SYSTEM DESIGN

7th Semester and 4th Year

Program Elective - IV

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Design of Solar Thermal System	20
2.	Design of Wind Energy System	15
3.	Design of Biogas Plant	05
4.	Hybrid system	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1. Design of Solar Thermal System 20 Hours 45%
 - 1.1 Solar Radiation , Solar economics, Photovoltaic, Flat Plate Solar Collectors, Passive Solar Heating Systems, Active Solar Heating Systems, Solar Hot Water Systems..
 - 1.2 Solar PV system Design.
2. Design of Wind Energy System 15 Hours 33%
 - 2.1 Wind data and energy estimation – Wind energy Conversion systems – Wind energy generators and performance estimation of Wind Turbine power rating.

3 **Design of Biogas Plant** **05 Hours** **11%**

3.1 Introduction to biomass, Bio-gas as a source of energy, types of Biogas plant.

3.2 Design of Biogas plant

4 **Hybrid System** **05 Hours** **11%**

PV/Solar thermal/grid-connected hybrid System, Biomass-PV-Diesel Hybrid System,
Solar wind hybrid system

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behaviour will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Design various solar thermal systems.

CO2: Compare the performance of different solar collectors.

CO3: Estimate Wind Turbine power rating.

CO4: Design Bio-gas plant.

CO5: Understand different hybrid system.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	2	-	-	-	-	-	-	-	-	-	2	-
CO2	1	1	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	2	-	-	-	-	-	-	-	-	-	-	2	-
CO4	3	2	2	-	-	-	-	-	-	-	-	-	3	-
CO5	2	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books/Reference Books:

1. Domkundwar & Domkundwar “Solar Energy and Non-Conventional Energy sources”, Dhanpat Rai & Co., 2014.
2. Sukatame, “Solar Energy”, Tata McGraw Hill, New Delhi.
3. G. D. Rai, “Energy Sources” 2nd Ed. by, Khanna Publications, New Delhi.
4. Rao & Parulaker “Energy Technology”, Khanna Publishers

Reference Books:

1. Suneel Deambi, “Photovoltaic System Design: Procedures, Tools and Applications” CRC Press, 2016.
2. Mukund R. Patel, “Wind and Solar Power Systems: Design, Analysis, and Operation” CRC press, 2006.
3. Zhifeng Wang, “Design of Solar Thermal Power Plants”, Elsevier Science & Technology, 2018.

Web Materials:

1. <http://www.nptel.ac.in/>

Other Materials:

1. www.sciencedirect.com
2. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>
3. IEEE: www.ieeeexplore.ieee.org

ME – 477 OPTIMIZATION TECHNIQUES

7th Semester and 4th Year

Program Elective- IV

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Optimization	05
2.	Classical Optimization Techniques	15
3.	Non-Linear Programming: Unconstrained Optimization Techniques	25

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

- | | | | |
|-----|---|----------|-----|
| 1. | Introduction to Optimization | 05 Hours | 11% |
| 1.1 | Introduction | | |
| 1.2 | Classification of Optimization problems | | |
| 1.3 | Statement of an optimization problem, design vector, design constraints, constraint surfaces, objective function & surfaces | | |
| 1.4 | Requirements for the application of Optimization methods | | |
| 1.5 | Applications of Optimization in engineering | | |
| 2. | Classical Optimization Techniques | 15 Hours | 33% |
| 2.1 | Introduction, Single-Variable Optimization | | |
| 2.2 | Multivariable Optimization with No Constraints, Saddle Point | | |
| 2.3 | Equality & Inequality Constrained Problems: Lagrange Multipliers, Kuhn-Tucker Conditions | | |

3. Non-Linear Programming: Unconstrained Optimization Techniques 25 Hours 56%

3.1 Unimodal function

3.2 Functions of Single Variable:

Elimination methods like Unrestricted Search, Exhaustive Search, Dichotomous Search, Interval-halving Method, Fibonacci Method and Golden-section Method

Interpolation Methods, Direct Root Methods like Newton Method, Quasi-Newton Method and Secant Method

3.3 Functions of Several Variables: Direct Search Methods, Gradient-based Methods and Numerical Methods

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject, course outcomes, learning outcomes will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Explain various classical optimization techniques and their applications in engineering.

CO2: Develop optimization models for various engineering problems.

CO3: Identify the mathematical tools that are needed to solve optimization problems.

CO4: Solve optimization problems with the help of software.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	1	-	-	-	-	-	-	-	-	-	-	-
CO2	-	2	1	-	-	-	-	-	-	-	-	-	-	-
CO3	-	2	1	1	-	-	-	-	-	-	-	-	-	-
CO4	-	2	2	1	2	-	-	-	-	-	-	1	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Textbooks:

1. Rao Singaresu S, “Engineering Optimization: Theory and Practice”, New Age Intl. Ltd., Publishers.
2. Deb Kalyanamoy, “Optimization for Engineering design: algorithms and Examples”, Prentice Hall of India.
3. Reklaitis G V, Ravindran A and Ragsdell K M, “Engineering Optimization: Methods and Application”, Wiley.

Reference Books:

1. Arora J. S., “Introduction to Optimum Design”, Elsevier.
2. Unwubolu Godfrey C. and Babu B. V., “New Optimization Techniques in Engineering”, Springer.
3. Dennis J. Jr and Schnabel R., “Numerical Methods for Unconstrained Optimization and Nonlinear Equations”, Society for Industrial and Applied Mathematics.
4. Fox R. L., “Optimization Methods for Engineering Design”, Addison Wesley.

Web Material:

1. <http://nptel.iitm.ac.in>

Other Material:

1. Programming Languages: MATLAB
2. Science Direct Journal (<http://www.sciencedirect.com>)
3. IEEE transactions (<http://ieeexplore.ieee.org>)
4. Mechanical Engg. (Inst. of Engineers) <http://www.ieindia.org/publish/mc/mc.html>
5. Production Engg. (Inst. of Engineers) <http://www.ieindia.org/publish/pr/pr.html>

ME478 INDUSTRIAL TRIBOLOGY

7th Semester and 4th Year

Programme Elective -IV

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Tribology	05
2.	Friction	10
3.	Wear	10
4.	Lubrication and Lubricants	15
5.	Application of Tribology	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Introduction to Tribology	05 Hours	11%
1.1	Introduction to tribology		
1.2	History of tribology		
1.3	Interdisciplinary Approach		
1.4	Economic Benefits		
2	Friction	10 Hours	22%
2.1	Introduction to friction, Laws of friction, Causes of friction, theories of friction and measurements of friction.		

3 **Wear** 10 Hours 22%

3.1 Classification and Mechanisms of Wear, factor affecting of wear

3.2 Theories of wear and measurements of wear

3.3 Wear control mechanism methods

4 **Lubrication and Lubricants** 15 Hours 34%

4.1 Importance of Lubrication, Boundary Lubrication, Mixed Lubrication, Full Fluid Film Lubrication; Hydrodynamic, Elastohydrodynamic lubrication, Types & Properties of Lubricants, Lubricants Additives.

5 **Application of Tribology** 05 Hours 11%

5.1 Introduction, Rolling Contact Bearings, Gears, Journal Bearings - Finite Bearings.

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of this course, the student will be able to:

CO1: Identify the tribological properties for various industrial applications.

CO2: Analyze the effect of various parameters on friction and wear.

CO3: Interpret the various lubricants and its properties

CO4: Apply the knowledge of Industrial Tribology for various applications.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	-	-	-	3	-
CO2	2	2	-	-	-	-	-	-	-	-	-	-	2	-
CO3	3	3	-	-	-	-	-	-	-	-	-	-	3	-
CO4	3	3	-	-	-	-	-	-	-	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Phakatkar H.G., Ghorpade R.R., “Tribology”, Nirali Prakashan
2. Basu S.K., Sengupta S. N. , “Fundamental of Tribology” , PHI Learning Private Ltd.

Reference Books:

1. Sahoo Prasanta , “Engineering Tribology”, PHI Learning Pvt. Ltd
2. Srivatsava Sushil Kumar, “Tribology in Industry”, S. Chand and Co.
3. Tribological Design Data Guide, Part 1: Lubrication, 1995, The Institution of Mechanical Engineers, Tribology Group, UK.
4. Tribological Design Data Guide, Part 2: Lubrication, 1995, The Institution of Mechanical Engineers, Tribology Group, UK.
5. Cameron A, “Basic Lubrication Theory”, Longmans, 1971,
6. Hutchings I. M., “Tribology: Friction and Wear of Engineering Materials”, Edward Arnold Ltd, 1992.
7. Neale M.J., “Bearings: A Tribology Handbook”, Butterworth Heinemann, 1993.
8. Szeri A., “Tribology”, McGraw Hill Co., 1980, Taylor and Francis (reprint).
9. Williams J.A., “Engineering Tribology”, Oxford University Press, New York, 1994.

Other Materials:

1. www.tribology-abc.com ; www.ltu.se/tfm/me
2. Journal of Tribology, American Society of Mechanical Engineers.
3. www.skf.com ; www.stle.org
4. www.statoillubricants.com
5. Journal of Engineering Tribology, Mechanical Engineering Publications
6. www.imperial.ac.uk/tribology
7. <http://rotorlab.tamu.edu/me626>

ME479 SURFACE ENGINEERING

7th Semester and 4th Year

Programme Elective - IV

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

C. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Surface engineering	05
2.	Surface preparation	10
3.	Surface deposition techniques	15
4.	Characterization of coatings and surfaces	08
5.	Functional coatings and applications	07

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

D. Detailed Syllabus:

- 1 Surface engineering 05 Hours 11%
 - 1.1 Introduction to surface engineering and its terminology, classification of surface engineering processes, microstructure and properties.
 - 1.2 Current applications of surface engineering, frontier areas for applications of surface engineering, selection criteria for surface engineering processes.
- 2 Surface preparation 10 Hours 22%
 - 2.1 Introduction to surface preparation, external cleaning methods such as gross cleaning, specific cleaning and by applications of fluids.

- 2.2 Rinsing, drying, outgassing and outdiffusion.
- 2.3 Evaluating and monitoring of cleaning by various cleaning tests.
- 2.4 In situ cleaning, recontamination of surfaces in the ambient environment and in deposition systems.

3 Surface deposition techniques 15 Hours 34%

- 3.1 Introduction to Physical Vapor Deposition (PVD) processes, various types of PVD processes.
- 3.2 Evaporation process and apparatus, evaporation sources, thermal evaporation, electron beam evaporation.
- 3.3 Molecular Beam Epitaxy (MBE), ion plating, Pulsed Laser Deposition (PLD), Atomic Layer Deposition (ALD).
- 3.4 Introduction to Chemical Vapor Deposition (CVD) processes, important reaction zones in CVD, Classifications of CVD reactions, components of CVD systems, Types of CVD processes.

4 Characterization of coatings and surfaces 08 Hours 17%

- 4.1 Measurement of coatings thickness, Surface roughness, Surface microscopy and topography, Atomic force microscopy (AFM), scanning electron microscope (SEM).

5 Functional Coatings and Applications 07 Hours 16%

- 5.1 Functional and nano-structured coatings and their applications in photovoltaics, bio- and chemical sensors. Surface passivation of semiconductors & effect on electrical properties. Surface engineering of polymers and composites. Thin film technology for multilayers & superlattices for electronic, optical and magnetic devices

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of Multi-Media projector, Black Board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.

- Experiments/Tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.
- Industrial visits will be organized for students to explore industrial facilities. Students are required to prepare a report on industrial visit and submit as a part of the assignment.

D. Course Outcomes (COs):

On successful completion of this course, the student will be able to:

CO1: Prepare the substrate surface for coating

CO2: Develop the coatings on various materials

CO3: Analyze the surface of the coating

CO4: Identify the functional application of coating

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	2	-	-	-	-	-	-	-	1	-
CO2	2	-	-	-	3	-	-	-	-	-	-	-	2	-
CO3	2	1	-	-	2	-	-	-	-	-	-	-	2	-
CO4	3	2	-	-	-	-	-	-	-	-	-	-	-	-

E. Recommended Study Material:

Text Books/ Reference Books

1. P.M. Martin, "Handbook of Deposition Technologies for Films and Coatings: Science, Applications and Technology", Elsevier USA (2010).
2. M. Ohring, "The Materials Science of Thin Films", Academic Press Inc, San Diego, (1992).
3. D.M. Mattox, "Handbook of Physical Vapor Deposition (PVD) Processing, Film Formation, Adhesion, Surface Preparation and Contamination Control", Noyes Publications U.S.A., (1998).
4. H.Y. Erbil, "Surface Chemistry of Solid and Liquid Interfaces", Blackwell Publishing Ltd, UK, (2006).
5. Satyanarayana, V.N.T. Kuchibhatla, A.S.Karakoti, D. Bera, S. Seal, "One dimensional nanostructured materials", Progress in Materials Science 52 699–913 (2007).
6. S.C. Tjong, H. Chen, "Nanocrystalline materials and coatings", Materials Science and Engineering R 45 1–88 (2004).
7. B. Duncan, R. Mera, D. Leatherdale, M. Taylor, R. Musgrove, "Techniques for characterising the

wetting, coating and spreading of adhesives on surfaces", NPL Report DEPC MPR 020 1-42 (2005

8. A.I. Gusev, A.A. Rempel, "Nanocrystalline Materials", Cambridge International Science Publishing UK, (2004).
9. G. Cao, "Nanostructures & Nanomaterials: Synthesis, Properties & Applications", Imperial College Press London, (2004).
10. B. Bhushan, "Springer Handbook of Nanotechnology", Springer Berlin Heidelberg New York USA, (2004).

Other Materials:

1. http://nptel.iitm.ac.in/courses/Webcoursecontents/IIT%20Kharagpur/Manuf%20Proc%20II/New_index1.html
2. <http://www.sciencedirect.com/science/journal>
3. <http://www.ieindia.org/publish/pr/pr.htm>
4. <http://www.ieindia.info/public.asp/me/me.htm>
5. <http://www.ias.ac.in/sadhana>

ME480 ADVANCED MANUFACTURING TECHNOLOGY

7th Semester and 4th Year

Program Elective- IV

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Advanced Machining Processes	16
2.	Thin Film Coating and Deposition Processes	08
3.	Additive Manufacturing	05
4.	Composites Manufacturing	06
5.	Advanced Metrology and Material characterization techniques	06
6.	Digital Manufacturing	04

Total hours (Theory): 45

Total hours (Lab): 30

Total: 75

B. Detailed Syllabus:

- 1 Advanced Machining Processes 16 hours 35%
- 1.1 Mechanical energy based advanced machining processes like ultrasonic machining, abrasive jet machining-process parametric analysis, process capabilities, and applications.
- 1.2 Thermoelectric based advanced machining processes like electro discharge machining, wire EDM, plasma arc machining, laser beam machining, focused ion beam machining-working principles, material removal mechanisms, process capabilities and applications.

- 1.3 Electrochemical and chemical based machining-electrochemical machining, electrochemical grinding, chemical machining-process characteristics, applications and limitations.
- 1.4 Hybrid processes abrasive water jet machining-abrasive flow finishing-magnetic abrasive finishing-magnetorheological finishing-ElectroStream drilling, shaped tube electrolytic machining- introduction, working principles, process performance, application.
- 1.5 Micro and nano machining processes, advanced processes like explosive forming, water hammer forming etc.

2 Thin Film Coating and Deposition Processes 08 hours 18%

- 2.1 Metal and inorganic material coatings.
- 2.2 Vapor deposition processes like physical vapor deposition (PVD), chemical vapor deposition (CVD) and its variants.
- 2.3 Molecular beam epitaxy (MBE), ion plating, pulsed laser Deposition (PLD), atomic layer deposition (ALD).
- 2.4 Introduction to sputtering, generation of energetic particles, sputtering yield.
- 2.5 Types of sputtering processes such as DC sputtering, RF sputtering, magnetron sputtering, reactive magnetron sputtering, target poisoning.

3 Additive Manufacturing (AM) 06 hours 13%

- 3.1 Essentials of AM, AM process chains, AM data formats.
- 3.2 Various types of AM systems i.e., Stereo lithography. FDM, SLS, SLA, LOM, RFP etc.
- 3.3 Factors concerning to AM: consideration for adoptions, advantages, accuracy and economic consideration.

4 Composites Manufacturing 06 hours 13%

- 4.1 Introduction to composites: function of the matrix and reinforcement in composites, fiber reinforcement.
- 4.2 Various polymer composite manufacturing processes: lay-up processes, spray up process, fiber placement process, resin transfer moulding, vacuum assisted resin transfer moulding, compression molding process, filament winding, calendaring.
- 4.3 Metal matrix composites, solid state processing, liquid state processing.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	-	-	-	-	-	-	1	1	-	2	-	-
CO2	3	2	2	2	-	-	-	-	-	-	-	-	2	-
CO3	1	1	-	-	-	-	-	-	-	-	-	-	-	-
CO4	2	2	-	-	2	-	-	-	-	-	-	2	2	-
CO5	1	2	1	-	-	-	-	-	-	-	-	1	-	-
CO6	-	-	1	-	1	-	1	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books/ Reference books:

1. Jain V. K., “Advanced Machining Processes”, Allied Publishers, Mumbai
2. Benedict. G.F. “Nontraditional Manufacturing Processes” Marcel Dekker Inc., New York
3. Pandey P.C. and Shan H.S. “Modern Machining Processes” Tata McGraw-Hill, New Delhi
4. Jain V. K., “Introduction to Micro machining”, Narosa publication
5. P.M. Martin, "Handbook of Deposition Technologies for Films and Coatings: Science, Applications and Technology", Elsevier USA (2010)
6. M. Ohring, "The Materials Science of Thin Films", Academic Press Inc, San Diego, (1992)
7. Pham D.T. and Dimov S.S, “Rapid Manufacturing”, Springer-Verlag, London, (2001)
8. Ghosh A., “Rapid Prototyping: A Brief Introduction”, East West Press, (2003)
9. Raghavendra N. V., Krishnamurthy L., “Engineering Metrology and Measurements”, Oxford University Press (2013)
10. Douglas B. Murphy,” Fundamentals of light microscopy and electronic imaging”, Wiley-Liss, Inc. USA, (2001)
11. Gilchrist A.,”Industry 4.0: The industrial internet of things”, Apress, (2016)
12. Zude Z., Shane X., Dejun C. “Fundamentals of Digital Manufacturing Science”, Springer (2012)

ME481 COMPUTATIONAL FLUID DYNAMICS

7th Semester and 4th Year

Program Elective- IV

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Computational Fluid Dynamics	10
2.	Finite Difference Method	15
3.	Finite Volume Method	10
4.	Finite Element Analysis	05
5.	Discretization of Navier-Stokes Equations	05

Total hours (Theory): 45

Total hours (Lab): 30

Total: 75

B. Detailed Syllabus:

1. Introduction to Computational Fluid Dynamics 10 Hours 22%
 - 1.1 Introduction, Need of Computational Fluid Dynamics. Applications of CFD.
 - 1.2 Introduction of Software's for CFD.
 - 1.3 Conservation equation; mass; momentum and energy equations; convective forms of the equations and general description.
 - 1.4 Mathematical Behaviour of Partial Differential Equations.
 - 1.5 Grid Generation
 - 1.6 Discretization Approaches – Finite Element Method, Finite Difference Method, Finite Volume Method

- | | | |
|--------------|---|----------------------------|
| 2 | Finite Difference Method | 15 Hours 34% |
| 2.1 | Introduction to Finite Differences, Taylor's expansion series, approximation techniques for first order derivative, second order derivative and mixed derivative for uniform and non-uniform grid, using forward difference scheme, backward difference scheme and central difference scheme, Steady state one-dimensional conduction equation. | |
| 2.2 | Discretization, Steady state one dimensional conduction and convection equation analytical solution and numerical solution using forward difference, backward difference and the central difference schemes | |
| 2.3 | Discretization of time dependent parameter using explicit scheme, implicit scheme, Errors and Analysis of stability. | |
|
3 |
Finite Volume Method |
10 Hours 22% |
| | Introduction to Finite Volume Method: Integral approach, Discretization and higher order schemes, 1-D Steady State Diffusion Problems. | |
|
4 |
Finite Element Method |
05 Hours 11% |
| 4.1 | Introduction to Finite Element Method: Basics of finite element method, stiffness matrix, isoperimetric elements, formulation of finite elements for one- and two-dimensional elements; applications. | |
| 4.2 | Finite Element Analysis in different Engineering applications like Heat Transfer and Fluid flow. | |
|
5 |
Discretization of Navier-Stokes Equations |
05 Hours 11% |
| | Stream Function Vorticity approach and Primitive variable approach, SIMPLE Algorithm, SIMPLER Algorithm. | |

C. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.

- In the lectures behaviour will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Describe various governing equations applicable for viscous fluid flows.

CO2: Describe the mathematical behavior of partial differential equations.

CO3: Apply Finite Difference Method to one dimensional and two dimensional Engineering problems.

CO4: Formulate and solve fluid flows and Heat transfer problems by Finite Element Method and Finite Volume Method.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	3	3	-	2	1	-	-	-	-	-	-	-	3	-
CO4	3	3	3	2	3	-	-	-	-	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Patankar, “Numerical heat transfer and Fluid Flow”, Mc.GrawHill. 2002.
2. John D Anderson Jr., “Computational Fluid Dynamics: The Basics with Applications”, McGraw-Hill, 1995.
3. Chandrupatla and Belegundu, “Introduction to Finite Element Methods”, Prentice Hall of India.
4. Peric and Ferziger, “Computational Methods for Fluid Dynamics”, Springer Publication.
5. Veersteeg and Malalasekara, “CFD: The Finite Volume Method by”, Prentice Hall, 1996.

Reference books:

1. Carnahan B, “Applied numerical method” John Wiley and Sons-2001.
2. Dante A W, “Introduction to Computational Fluid Dynamics”, Cambridge Uni. Press, 2005.

3. Reddy J N and Gartling D K, "The Finite Element Method in Heat Transfer and Fluid Dynamics", CRC Press, 2000.
4. White F M, "Viscous Fluid Flow", Mc Graw Hill, 1991.
5. Schlichting H, Gersten K, "Boundary-Layer Theory", 8th edition, 2004.
6. Anderson, J D, "Modern Compressible Flow: With Historical Perspective", McGrawHill, 2002.
7. Murlidhar K and Sunderrajan T, "Computational Fluid Mechanics and Heat Transfer", Narosa Publishing House.
8. Goshdastidar, "Computer Simulation of Flow and Heat Transfer", Tata-McGraw Hill.

Web Material:

1. <https://nptel.ac.in/courses/112/105/112105045/>
2. www.sciencedirect.com:
International Journal of Heat and Fluid Flow.
International Journal of Thermal Sciences.
Experimental Thermal and Fluid Science.

Other Materials:

1. Programming Languages and Software's: ANSYS, COMSOL.

ME494 INDUSTRIES 4.0 & INDUSTRIAL IOT

7th Semester and 4th Year

Program Elective- IV

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to Industry 4.0	05
2.	Main components of the industry 4.0	10
3.	Industry 4.0 design principles	10
4.	Smart factory	10
5.	Standards of industry 4.0 and Safety and Security in networked Production Environments	10

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

B. Detailed Syllabus:

1	Introduction to Industry 4.0	07 Hours	10%
1.1	Industry 1.0 TO 3.0(Evolution of Industry 1.0)		
1.2	Meaning of Industry 4.0(Upgradation to Industry 4.0)		
1.3	Comparison of Industry 4.0 Factory and <u>today's Factory</u> (Older Factory)		
1.4	Difference between conventional automation and Industry 4.0		
2	Main components of the industry 4.0	07 Hours	10%
2.1	Cyber-Physical Systems,		
2.2	Internet of Things		

2.3	Smart Factory and Smart Machines		
2.4	Internate of services(SAAS – Software As A Service)		
3	Industry 4.0 design principles	06 Hours	10%
3.1	Horizontal integration		
3.2	Vertical integration		
4	Smart factory	08 Hours	12%
4.1	Steps to a smart factory		
4.2	Basic principles and technologies of a Smart Factory		
5	Standards of industry 4.0 and Safety and Security in networked Production Environments	04 Hours	10%
5.1	Success with standards		
5.2	Road map of standards.		
5.3	Case studies approaching industry 4.0		
5.4	Safety for connected Machines and Systems		
5.5	Safety in Human Robot cooperation		

C. Instructional Method and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures behavior will be observed strictly.

D. Students Learning Outcomes:

On successful completion of the course, the student will be able to:

CO1: Describe the concept of internet of Things with respect to earlier revolutions.

CO2: Identify the main components of industry 4.0

CO3: Explain design principles of industry 4.0 and internet of things.

CO4: Explain the basic principles for industry 4.0. for smart factory.

CO5: Explain various standards and safety issues of Industry 4.0.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	2	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO3	-	-	-	-	-	-	-	-	-	-	-	-	2	-
CO4	-	-	-	-	-	-	-	-	-	-	-	-	2	-
CO5	-	2	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Gilchrist, Alasdair “Industry 4.0,” 1st Edition.2016
2. Madiseti, ArshdeepBahgavijay, “Internet of Things,” 1st Edition.2014
3. Dr. SRN Reddy, RachitThukral and Manasi Mishra, “Introduction to Internet of Things: A practical Approach”, ETI Labs
4. Cuno Pfister, “Getting Started with the Internet of Things”, O Reilly Media Maker Media, Inc; 1st edition 2011
5. Alp Ustundag and Emre Cevikcan,”Industry 4.0: Managing the Digital Transformation”.
6. Bartodziej, Christoph Jan,”The Concept Industry 4.0”.

Reference Books:

1. Pethuru Raj and Anupama C. Raman, “The Internet of Things: Enabling Technologies, Platforms, and Use Cases”, CRC Press
2. Jeeva Jose, “Internet of Things”, Khanna Publishing House New Delhi; 1st edition , 2018
3. Klaus Schwab,”The Fourth Industrial Revolution”. 2017
4. Christian Schröder ,”The Challenges of Industry 4.0 for Small and Medium-sized Enterprises”. Publisher: Friedrich-Ebert-Stiftung 2017

Web Material:

1. <https://courses.edx.org/courses/course-v1:HKPolyUx+I4.0x+1T2018/course/>
2. <https://www.coursera.org/specializations/digital-manufacturing-design-technology>
3. https://onlinecourses.nptel.ac.in/noc20_cs69/preview

B. Tech. (Mechanical Engineering) Programme

SYLLABI

(Semester – 8)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ME456 MAJOR PROJECT

8th Semester and 4th Year

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	28	28	14
Marks	-	400	400	

A. Outline of the Course:

- Major Project showcases that, a student during the final semester will be able to apply the engineering knowledge to a practical problem.
- Students can carry out their project work in-house at institute/ industry/ research organization/govt. organization.
- A student is required to carry out the project work related to Mechanical Engineering or some interdisciplinary work under the guidance of a faculty member and the supervisor of the concerned industry/institute/organization.
- The major project work provides student an opportunity to develop something on their own or group of maximum 3 students. Final decision related to the number of students in one group is based on internal supervisor recommendation.
- Project will be evaluated at least twice during the semester by internal faculty as a part of continuous evaluation and final evaluation will be carried out by external examiner at the end of semester.
- The project should be related to the Design, analysis and/or fabrication, Experimentation, Simulations and evaluation, Product design and development, Industry needs based basic survey or Testing or Analysis etc. Physical prototype development including detail design.
- Student should submit the project report at the end of the semester duly signed by project supervisor and head of department. Report comprising literature review, objective, methodology, scope of the project work undertaken, outcome of project and conclusions.
- The students are required to identify their problem and they are required to follow all the rules and instruction issued by department/industry.

B. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

C. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

1. Apply the engineering knowledge to analyze, provide solution and carried out investigation of the problems by individual or team.
2. Identify, analyze use the specific/modern tools/technology to achieve the solution
3. Analyze the problem in commercial, societal, health, safety, legal, cultural and environment aspects before reach to final solution.
4. Create plan and manage financial activities, documentation and team work.
5. Use professional ethical principles and responsibility during their project.
6. Demonstrate the soft skill during project work.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	-	-	-	-	-	-	-	-	-	3	3
CO2	-	3	3	3	-	-	-	-	-	-	-	-	3	3
CO3	-	-	3	3	3	3	3	3	-	-	-	-	3	3
CO4	-	-	-	-	-	-	-	-	3	-	3	2	1	1
CO5	-	-	-	-	-	-	-	2	-	3	-	2	1	1
CO6	-	-	-	-	-	-	-	-	-	3	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”